



$$y(t) = t \cdot x(t)$$

$$x^*(t) \rightarrow \boxed{} \rightarrow y^*(t)$$

Is this system additive?

$$\begin{aligned} T[x_1(t) + x_2(t)] &= t \cdot (x^*(t)) \\ &= t \cdot (x_1(t) + x_2(t)) \quad \text{--- (1)} \end{aligned}$$

$$\begin{aligned} T[x_1(t)] &= t x_1(t) = y_1(t) \\ T[x_2(t)] &= t x_2(t) = y_2(t) \end{aligned}$$

$$y_1(t) + y_2(t) = t(x_1(t) + x_2(t)) \quad \text{--- (11)}$$

Clearly R.H.S. of Eqⁿ (1) and Eqⁿ (11) are same, so they satisfy additivity property.

$$\textcircled{2} \quad y(t) = (x(t))^2$$

$$\begin{aligned} T[x_1(t) + x_2(t)] &= (x_1(t) + x_2(t))^2 \\ &= (x_1(t))^2 + (x_2(t))^2 + 2x_1(t)x_2(t) \end{aligned}$$

③

$$T[x_1(t)] = (x_1(t))^2 = y_1(t)$$

$$T[x_2(t)] = (x_2(t))^2 = y_2(t)$$

$$y_1(t) + y_2(t) = (x_1(t))^2 + (x_2(t))^2 \quad \text{--- (iv)}$$

Clearly R.H.S. of (iii) & (iv) are not equal.
So they are not additive.

③ $y(t) = \sin(x(t))$

Is this system Homogenous?

$$T[\alpha x(t)] = \sin(\alpha x(t)) \quad \text{--- (i)}$$

$$\alpha T[x(t)] = \alpha \sin(x(t)) \quad \text{--- (ii)}$$

Eqⁿ (i) and Eqⁿ (ii) have different R.H.S.
So they are not homogenous.

④ $y(t) = t \cdot x(t)$

Let the $x_d(t) = x(t - \tau)$

$$T[x_d(t)] = t \cdot x(t - \tau) \quad \text{--- (i)}$$

$$y(t-\tau) = (t-\tau) \cdot x(t-\tau) \quad \text{--- (ii)}$$

Clearly, R.H.S of eqⁿ (i) and eqⁿ (ii) are not the same.

So this is not time-invariant system.

Fundamental Time period of Sum of Signals.

The fundamental time period of a sum of periodic signals is the Least Common Multiple (LCM) of the individual periods, provided the ratio of any two periods is rational.

Step 1:

- ① Find the fundamental period of each individual signal.
- ② Check if their ratio is rational
- ③ If yes to ② then we find LCM by
 - (i) Express each period as a fraction.
 - (ii) Find LCM of the numerators and GCD (Greatest Common divisor) of the denominator.

$$\text{LCM} = \frac{\text{LCM}(\text{numerators})}{\text{GCD}(\text{denominators})}$$

Example.

① Find the fundamental period of $x(t) = \cos(\underline{4\pi t}) + \sin(\underline{6\pi t})$

$$T_1 = \frac{2\pi}{4\pi} = 0.5 \text{ s} = \frac{1}{2}.$$

$$T_2 = \frac{2\pi}{6\pi} = \frac{1}{3} \text{ s}.$$

$$\frac{T_1}{T_2} = \frac{1}{2} \cdot \frac{3}{1} = \frac{3}{2} = 1.5 \text{ which is rational.}$$

L.C.M of 1 and 1 is 1

G.C.D of 2 and 3 is 1

$$\text{so } T = \frac{1}{1} = 1$$

so fundamental period is 1.

② $x(t) = \cos(2\pi t) + \sin(3\pi t) + \cos(5\pi t) + \sin(7\pi t)$

$$T_1 = \frac{2\pi}{2\pi} = 1 \text{ s} = \frac{105}{105}$$

$$T_2 = \frac{2\pi}{3\pi} = \frac{2}{3} \text{ s} = \frac{70}{105} \quad \text{first denote everything as}$$

$$T_3 = \frac{2\pi}{5\pi} = \frac{2}{5} \text{ s} = \frac{42}{105} \quad \text{same denominator}$$

$$T_4 = \frac{2\pi}{7\pi} = \frac{2}{7} \text{ s} = \frac{30}{105} \quad 7 \times 5 \times 3 = 105$$

$$\begin{aligned} \text{LCM of numerators } (105, 70, 42, 30) \\ = 210 \end{aligned}$$

$$\text{LCM of numbers} = 210$$

$$\text{GCD} = 105$$

$$T = \frac{210}{105} = 2$$

Common fundamental Time period of the Sum is
2 seconds.